Notes: Jones and Kaul (JF, 1996)

Oil and the Stock Markets

Contents

1 Big picture

- **Question.** Do stock markets react rationally to oil price shocks? More precisely, can the response of stock prices to oil shocks be justified by changes in current and expected future real cash flows and/or by changes in expected stock returns?
- **Core setting.** The paper studies postwar quarterly data for four major stock markets: the United States, Canada, Japan, and the United Kingdom. These countries differ in oil dependence, industrial structure, and market institutions, so they provide useful cross-country evidence.
- **Why oil shocks are useful.** Oil is a major input into production and consumption. Large postwar oil price movements are plausibly driven by petroleum-sector events such as OPEC actions, wars, and geopolitical disruptions. This makes oil shocks a useful source of macroeconomic news whose real effects should be visible in output and cash flows.
- **Economic puzzle.** Oil price increases have negative effects on stock returns in all four countries. The key question is whether this negative stock-market response is "right-sized" relative to the later decline in real cash flows and to any change in expected returns.
- **Valuation framework.** The paper uses the Campbell-Shiller style log-linear return decomposition. A stock return can move because expected future cash flows change or because expected future returns change. If an oil shock affects prices only through these rational channels, then oil variables should have no residual explanatory power once cash-flow news and expected-return variables are included.
- **Main empirical result.** For the United States and Canada, oil shocks do not have residual explanatory power once current and future industrial production growth is included. This supports the view that these markets rationally evaluate the real effects of oil shocks.
- **Main contrast.** For Japan and the United Kingdom, oil shocks continue to explain stock returns even after conditioning on future real cash flows and standard expected-return proxies. The stock-price response appears too large relative to measured subsequent cash-flow effects.
- **Interpretation.** The authors offer two possible interpretations for Japan and the United Kingdom: either oil shocks move expected returns through channels not captured by their proxies, or those stock markets overreact to oil shocks.
- **Contribution.** The paper is not only about oil. It uses a macro shock with plausible real consequences to evaluate whether stock prices process economically important news rationally.
- **Takeaway.** In the United States and Canada, oil-price news appears to be translated into stock prices through real cash-flow fundamentals. In Japan and the United Kingdom, the same test leaves unexplained stock-price volatility.

2 Data and measurement (key objects)

2.1 Countries and sample periods

The analysis uses quarterly data. The sample periods differ because the authors require stock returns, inflation, industrial production, oil-price indices, and expected-return proxies for each country.

Country	Sample period used in the main tables
United States	1947–1991
Canada	1960–1991
Japan	1970–1991
United Kingdom	1962–1991

The U.S. sample covers the longest postwar window. Japan starts later, so the authors also run common 1970–1991 subsample checks to ensure that the cross-country conclusions are not simply driven by using different periods.

2.2 Real stock returns

The dependent variable is the real stock return:

$$\mathrm{RS}_t = S_t - I_t, \tag{1}$$

where S_t is the continuously compounded nominal return on the country's aggregate stock market index and I_t is inflation, measured using the log change in the consumer price index.

Interpretation. Oil shocks are real supply-side disturbances, so the relevant stock-market response is the real return, not just the nominal return. The authors later check that using nominal returns does not overturn the main conclusions.

2.3 Oil shocks

The main oil variable is the prewhitened percentage change in an oil-related producer price index:

$$\mathrm{OIL}_t = \text{innovation in the log oil-price index.} \tag{2}$$

The underlying producer price index differs slightly by country:

- United States: fuels and related products and power;
- Canada and Japan: petroleum and coal products;
- United Kingdom: fuel.

The authors prewhiten the oil-price changes to remove serial correlation that may arise because components of price indices are sampled nonsynchronously.

Interpretation. The goal is to use oil-price innovations, not predictable oil-price drift. If lagged oil changes predicted current stock returns merely because of mechanical index smoothing, the efficiency test would be contaminated.

2.4 Cash-flow proxy

The cash-flow proxy is industrial production growth:

$$\mathrm{IP}_t = \Delta \log(\text{seasonally adjusted industrial production index})_t. \tag{3}$$

This is an aggregate real-activity measure. The paper treats current and future industrial production growth as observable proxies for current and expected future real cash-flow changes.

Important measurement issue. Theory calls for expected future cash-flow growth, but the econometrician observes realized future industrial production growth. Realized future cash flow contains measurement error relative to the market's expectation at time t.

2.5 Expected-return proxies

The expected-return variables are standard predictors from the stock-return predictability literature:

$$X_{1t} = \mathrm{DY}_{t-1}, \qquad \text{dividend yield,} \tag{4}$$
$$X_{2t} = \mathrm{DEF}_{t-1}, \qquad \text{default spread,} \tag{5}$$
$$X_{3t} = \mathrm{TERM}_{t-1}, \qquad \text{term spread,} \tag{6}$$
$$X_{4t} = \mathrm{SDEF}_{t}, \qquad \text{shock to the default spread,} \tag{7}$$
$$X_{5t} = \mathrm{STERM}_{t}, \qquad \text{shock to the term spread.} \tag{8}$$

The default spread is the difference between corporate and long-term government bond yields. The term spread is the difference between long-term government bond yields and short-term Treasury bill yields. The shock variables are residuals from first-order autoregressive models for the corresponding spreads.

Interpretation. These variables are included because oil shocks may change required returns, not only expected cash flows. However, they are financial variables, so they may themselves be affected by valuation errors or market sentiment.

2.6 Granger-precedence of oil prices

The authors report Granger-causality tests showing that, with the exception of the United Kingdom, oil-price changes Granger-precede both stock returns and output. This supports the interpretation that oil shocks are useful macroeconomic news rather than merely a response to the same stock-market forces being explained.

Caution. Granger-precedence is not the same as a perfect natural experiment. It is a timing and predictability condition. The paper's credibility comes from combining that timing evidence with the real-economy importance of oil and robustness checks for measurement error.

3 Models and empirical specifications (all equations + interpretation)

3.1 Log-linear stock-return decomposition

The theoretical foundation is a log-linear valuation identity. The real stock return can be written as

$$\mathrm{RS}_t = E_{t-1}(\mathrm{RS}_t) + (E_t - E_{t-1}) \sum_{j=0}^{\infty} \rho^j \Delta C_{t+j} - (E_t - E_{t-1}) \sum_{j=1}^{\infty} \rho^j \mathrm{RS}_{t+j}, \tag{9}$$

where ΔC_{t+j} denotes real cash-flow growth and ρ is a discounting/log-linearization parameter close to one.

Interpretation. A stock price rises when the market receives good news about current or future cash flows. It falls when discount rates or required returns rise. Therefore oil shocks should matter for stock prices only through cash-flow news or expected-return news.

3.2 Residual oil-shock test

The empirical test can be summarized as

$$\mathrm{RS}_t = E_{t-1}(\mathrm{RS}_t) + (E_t - E_{t-1}) \sum_{j=0}^{\infty} \rho^j \Delta C_{t+j} - (E_t - E_{t-1}) \sum_{j=1}^{\infty} \rho^j \mathrm{RS}_{t+j} + \sum_{s=0}^{k} \theta_s \mathrm{OIL}_{t-s} + u_t. \tag{10}$$

The market-efficiency implication is

$$\theta_s = 0 \quad \text{for all } s. \tag{11}$$

Interpretation. If oil shocks are fully incorporated through cash-flow and expected-return channels, the oil variables should not remain significant after those channels are controlled for.

3.3 Step 1: do stock returns reflect future real cash flows?

The first empirical relation is

$$\mathrm{RS}_t = \alpha_1 + \sum_{s=0}^{4} \beta_{1s} \mathrm{IP}_{t+s} + \eta_{1t}. \tag{12}$$

Purpose. This regression checks whether stock returns are related to current and future real activity. A positive relation is necessary for the cash-flow test to make sense.

Key result. Stock returns are positively related to current and future industrial production growth in all four countries.

3.4 Step 2: do oil shocks affect stock returns?

The oil-only regression is

$$\mathrm{RS}_t = \alpha_2 + \sum_{s=0}^{4} \theta_{1s} \mathrm{OIL}_{t-s} + \eta_{2t}. \tag{13}$$

Purpose. This establishes whether oil shocks matter for stock returns before asking whether cash flows can explain that effect.

Key result. Oil price increases have negative effects on stock returns in all four countries. The strongest explanatory power appears for Japan.

3.5 Step 3: condition on real cash flows

The main cash-flow specification is

$$\mathrm{RS}_t = \alpha_3 + \sum_{s=0}^{4} \theta_{2s} \mathrm{OIL}_{t-s} + \sum_{s=0}^{4} \beta_{2s} \mathrm{IP}_{t+s} + \eta_{3t}. \tag{14}$$

The hypothesis test is again

$$H_0 : \theta_{2s} = 0 \quad \text{for all } s. \tag{15}$$

Interpretation. This is the central test under constant expected returns. If the oil coefficients disappear after adding current and future industrial production growth, the stock market response is consistent with rational cash-flow valuation.

3.6 Step 4: condition on real cash flows and expected-return variables

The most complete main regression is

$$\mathrm{RS}_t = \alpha_4 + \sum_{i=1}^{5} a_i X_{it} + \sum_{s=0}^{4} \theta_s \mathrm{OIL}_{t-s} + \sum_{s=0}^{4} \beta_s \mathrm{IP}_{t+s} + \eta_{4t}. \tag{16}$$

Purpose. This regression allows oil shocks to affect stock prices through both real cash flows and expected returns.

Interpretation. If oil coefficients remain significant here, the remaining stock-market reaction is hard to explain using standard rational valuation channels.

3.7 Measurement-error correction

Because expected cash flows are unobserved, realized future IP_{t+s} may be noisy. The authors therefore estimate augmented regressions that include future stock returns as proxies for the measurement error in realized future cash flows:

$$\mathrm{RS}_t = \alpha_3 + \sum_{s=0}^{4} \theta_{2s} \mathrm{OIL}_{t-s} + \sum_{s=0}^{4} \beta_{2s} \mathrm{IP}_{t+s} + \sum_{s=1}^{4} \gamma_{2s} \mathrm{RS}_{t+s} + \eta_{3t}, \tag{17}$$

with an analogous version that also includes the X_{it} expected-return variables.

Interpretation. If the residual oil effects in Japan and the United Kingdom were purely a by-product of noisy cash-flow measures, adding future returns should weaken or remove them. The authors report that it does not.

4 Identification, credibility, and robustness

4.1 What identifies the test

The paper's identification logic is not a randomized experiment. It is a valuation restriction:

$$\text{oil shock} \Rightarrow \text{cash-flow news and/or expected-return news} \Rightarrow \text{stock return.} \tag{18}$$

If the middle channels are controlled for, the direct oil-to-return relation should vanish in a rational pricing model. The test is powerful because oil shocks are economically important and tend to occur before, rather than after, stock-return and output movements.

4.2 Why oil shocks are credible news shocks

Oil shocks are attractive because they are large, observable, and plausibly tied to petroleum-sector events. Postwar oil price jumps include episodes linked to OPEC behavior, the 1973–1974 and 1979–1980 oil shocks, and the 1990 Iraq-Kuwait shock.

Economic intuition. If oil becomes more expensive, production costs rise, real output can fall, and corporate cash flows can decline. A rational stock market should discount those lower cash flows immediately.

4.3 Why real cash-flow controls matter

Many stock-market efficiency tests use only financial variables. Jones and Kaul emphasize that financial variables may themselves reflect fads or overreaction. Industrial production growth is a real macroeconomic variable, so it is less likely to be contaminated by the same stock-market sentiment being tested.

Implication. If real cash-flow controls alone eliminate oil-shock effects, that is relatively strong evidence for rational pricing. This is what happens in the United States and Canada.

4.4 Why expected-return controls are harder to interpret

Dividend yields, default spreads, and term spreads are useful predictors of expected returns, but they are financial variables. If stock prices are affected by sentiment, these expected-return proxies may also be affected by sentiment.

Implication. If the expected-return controls explain away oil effects, that evidence would be somewhat less clean than real cash flows alone. If even these controls fail, as in Japan and the United Kingdom, the puzzle becomes stronger.

4.5 Measurement errors in inflation

Because real stock returns subtract CPI inflation from nominal stock returns, inflation measurement error could matter. The authors reestimate the analysis using nominal stock returns. The measured oil effects and cash-flow relations are attenuated, but the main cross-country conclusions remain unchanged.

4.6 Measurement errors in oil prices

The oil variables are based on producer price indices, which may contain sampling errors. Classical measurement error in oil shocks would tend to bias oil coefficients toward zero. Therefore, the significant oil effects in Japan and the United Kingdom are unlikely to be caused simply by oil-price measurement error.

4.7 Measurement errors in cash-flow proxies

The main threat is that realized future industrial production growth is an imperfect proxy for expected future cash-flow news. The authors address this in two ways:

- they augment the regressions with future stock returns as proxies for cash-flow measurement error;
- they calculate how large cash-flow measurement error would have to be to explain the residual oil effects in Japan and the United Kingdom.

The required noise-to-signal ratios are about 1.38 for Japan and 1.22 for the United Kingdom. This implies that roughly 55–60% of realized real cash-flow variation would have to be unexpected noise, which the authors view as implausibly large.

4.8 Subperiod robustness

The sample periods differ by country. To address this, the authors repeat the analysis for a common 1970–1991 window and also estimate systems using seemingly unrelated regressions. The qualitative pattern remains: U.S. and Canadian results are consistent with rational cash-flow pricing, while Japan and the United Kingdom remain difficult to explain.

4.9 Main limitation

The paper cannot fully rule out the possibility that Japan and the United Kingdom have expected-return channels not captured by dividend yields, default spreads, term spreads, and spread shocks. Therefore the conclusion is not that these markets are definitively irrational; rather, the observed response is not explained by the rational channels the paper measures.

5 Tables and figures

5.1 Figure 1: Energy and petroleum expenditures in U.S. GNP

Read. Figure 1 shows that energy expenditures were a nontrivial share of U.S. gross national product during 1970–1988. Energy expenditures peaked around the early 1980s, and petroleum products alone also represented a meaningful share of GNP.

Economic message. Oil is not a minor input. A large oil-price shock can move production costs,

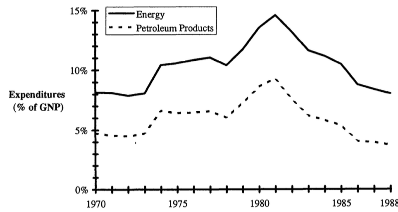

Figure 1. Share of Energy and Related Products in Gross National Product for the United States, 1970–1988. Annual U.S. energy expenditures (solid line) and expenditures on petroleum products (dashed line) as proportions of the gross national product.

household purchasing power, output, and therefore equity values.

5.2 Table I: Real stock returns and real cash flows

Table I estimates

$$\mathrm{RS}_t = \alpha_1 + \sum_{s=0}^{4} \beta_{1s}\mathrm{IP}_{t+s} + \eta_{1t}. \tag{19}$$

Country	$\sum_{s=0}^{4} \beta_{1s}$	$\bar{R}^2$	p-value for individual β test
United States	1.835	0.174	0.000
Canada	1.853	0.174	0.000
Japan	2.618	0.180	0.001
United Kingdom	3.730	0.104	0.005

Read. The coefficient sums are positive in all countries. The p-values show that current and future industrial production growth significantly explain real stock returns.

Economic message. Stock returns contain information about real cash flows. This supports the paper's strategy of asking whether oil-shock effects are absorbed by future real activity.

Table I

Regression Analysis of the Relation Between Real Stock Returns and Real Cash Flows Using Quarterly Data for the United States, Canada, Japan, and the United Kingdom

This table contains estimates of regression (4):

$$\mathrm{RS}_t = \alpha_1 + \sum_{s=0}^{4} \beta_{1s}\mathrm{IP}_{t+s} + \eta_{1t}$$

where RS_t = real stock return in quarter t measured as the difference between the continuously compounded nominal stock return, S_t, calculated from a country's stock market index and the inflation rate, I_t, computed as the logarithmic difference in the consumer price index (CPI); and IP_t = growth rate of industrial production in quarter t calculated as the logarithmic difference in the (seasonally adjusted) index of industrial production. The hypotheses tests, H_1 and H_2, are conducted using standard F-statistics; the p-values of these statistics are reported in parentheses.

Country	α_1	$\sum_{s=0}^{4} \beta_{1s}$	$\bar{R}^2$	H_1: $\sum_{s=0}^{4} \beta_{1s} = 0$	H_2: $\beta_{1s} = 0\ \forall s$
United States	−0.008	1.835	0.174	21.651	8.158
(1947–1991)				(0.000)	(0.000)
Canada	−0.016	1.853	0.174	12.588	5.961
(1960–1991)				(0.001)	(0.000)
Japan:	−0.008	2.618	0.180	14.618	4.476
(1970–1991)				(0.000)	(0.001)
United Kingdom:	−0.013	3.730	0.104	13.925	3.585
(1962–1991)				(0.000)	(0.005)

Table II

Regression Analysis of the Effects of Oil Shocks on Real Stock Returns Using Quarterly Data for the United States, Canada, Japan, and the United Kingdom

This table contains estimates of regression (5):

$$\mathrm{RS}_t = \alpha_2 + \sum_{s=0}^{4} \theta_{1s}\mathrm{OIL}_{t-s} + \eta_{2t}$$

where RS_t = real stock return in quarter t measured as the difference between the continuously compounded nominal stock return, S_t, calculated from a country's stock market index and the inflation rate, I_t, computed as the logarithmic difference in the consumer price index (CPI); and OIL_t = prewhitened percentage change in the oil price in quarter t computed as the logarithmic difference in the producer price indices for fuels and related products and power for the United States, for petroleum and coal products for Canada and Japan, and for fuel for the United Kingdom. All hypotheses tests, H_1 through H_4, are conducted using standard F-statistics; the p-values of these statistics are reported in parentheses.

Country	α_2	$\sum_{s=0}^{4} \theta_{1s}$	$\bar{R}^2$	H_1: $\sum_{s=0}^{4} \theta_{1s}=0$	H_2: $\sum_{s=1}^{4} \theta_{1s}=0$	H_3: $\theta_{1s} = 0\ \forall s$	H_4: $\theta_{1s} = 0\ \forall s > 0$
United States	0.008	−1.009	0.069	12.323	7.732	3.529	2.905
(1947–1991)				(0.001)	(0.006)	(0.005)	(0.023)
Canada	0.002	−0.992	0.028	6.911	6.434	1.672	1.904
(1960–1991)				(0.009)	(0.013)	(0.147)	(0.115)
Japan	0.019	−1.531	0.260	17.475	8.985	6.563	6.352
(1970–1991)				(0.000)	(0.004)	(0.000)	(0.000)
United Kingdom	0.003	−1.107	0.122	5.366	0.327	4.071	0.948
(1962–1991)				(0.023)	(0.569)	(0.002)	(0.439)

5.3 Table II: Oil shocks and real stock returns

Table II estimates

$$\mathrm{RS}_t = \alpha_2 + \sum_{s=0}^{4} \theta_{1s}\mathrm{OIL}_{t-s} + \eta_{2t}. \tag{20}$$

Country	$\sum_{s=0}^{4} \theta_{1s}$	$\bar{R}^2$	p-value for individual θ test
United States	-1.009	0.069	0.005
Canada	-0.992	0.028	0.147
Japan	-1.531	0.260	0.000
United Kingdom	-1.107	0.122	0.002

Read. Oil shocks have negative cumulative effects on stock returns in all four countries. Japan has by far the largest explanatory power: oil variables alone explain about 26% of real stock-return variation.

Economic message. Oil shocks matter for equity markets. The question is whether this is because the stock market correctly anticipates lower real cash flows, or because it overreacts.

5.4 Table III: Oil shocks conditional on cash flows

Table III estimates

$$\mathrm{RS}_t = \alpha_3 + \sum_{s=0}^{4} \theta_{2s}\mathrm{OIL}_{t-s} + \sum_{s=0}^{4} \beta_{2s}\mathrm{IP}_{t+s} + \eta_{3t}. \tag{21}$$

Country	$\sum \theta_{2s}$	$\sum \beta_{2s}$	$\bar{R}^2$	p-value: $\sum \theta_{2s} = 0$	p-value: $\theta_{2s} = 0\ \forall s$	p-value: $\beta_{2s} = 0\ \forall s$
United States	-0.465	1.559	0.177	0.123	0.383	0.000
Canada	-0.317	1.612	0.154	0.427	0.801	0.001
Japan	-1.042	1.719	0.328	0.016	0.002	0.040
United Kingdom	-0.606	2.892	0.189	0.242	0.010	0.022

Read. In the United States and Canada, oil coefficients become statistically insignificant after adding cash-flow controls. In Japan, oil remains significant by both the coefficient-sum test and the individual-coefficient test. In the United Kingdom, the sum test is insignificant, but the individual oil coefficients remain significant.

Economic message. U.S. and Canadian stock-market responses are consistent with cash-flow fundamentals. Japan and the United Kingdom still show residual oil-related stock-return movements after measured real cash flows are included.

5.5 Table IV: Oil shocks conditional on cash flows and expected returns

Table IV estimates

$$\mathrm{RS}_t = \alpha_4 + \sum_{i=1}^{5} a_i X_{it} + \sum_{s=0}^{4} \theta_s\mathrm{OIL}_{t-s} + \sum_{s=0}^{4} \beta_s\mathrm{IP}_{t+s} + \eta_{4t}. \tag{22}$$

Table III

Regression Analysis of the Effects of Oil Shocks on Real Stock Returns Conditional on Cash Flows Using Quarterly Data for the United States, Canada, Japan, and the United Kingdom

This table contains estimates of regression (6):

$$\mathrm{RS}_t = \alpha_3 + \sum_{s=0}^{4} \theta_{2s}\mathrm{OIL}_{t-s} + \sum_{s=0}^{4} \beta_{2s}\mathrm{IP}_{t+s} + \eta_{3t}$$

where RS_t = real stock return in quarter t measured as the difference between the continuously compounded nominal stock return, S_t, calculated from a country's stock market index and the inflation rate, I_t, computed as the logarithmic difference in the Consumer Price Index (CPI); IP_t = growth rate of industrial production in quarter t calculated as the logarithmic difference in the (seasonally adjusted) index of industrial production; and OIL_t = prewhitened percent change in the oil price in quarter t computed as the logarithmic difference in the producer price indices for fuels and related products and power for the United States, for petroleum and coal products for Canada and Japan, and for fuel for the United Kingdom. All hypotheses tests, H_1 through H_6, are conducted using standard F-statistics; the p-values of these statistics are reported in parentheses.

Country	α_3	$\sum_{s=0}^{4} \theta_{2s}$	$\sum_{s=0}^{4} \beta_{2s}$	$\bar{R}^2$	H_1: $\sum_{s=0}^{4} \theta_{2s}=0$	H_2: $\sum_{s=1}^{4} \theta_{2s}=0$	H_3: $\theta_{2s} = 0\ \forall s$	H_4: $\theta_{2s} = 0\ \forall s > 0$	H_5: $\sum_{s=0}^{4} \beta_{2s}= 0$	H_6: $\beta_{2s} = 0\ \forall s$
United States	−0.006	−0.465	1.559	0.177	2.401	0.768	1.132	0.711	12.765	5.331
(1947–1991)					(0.123)	(0.327)	(0.338)	(0.586)	(0.001)	(0.000)
Canada	−0.013	−0.317	1.612	0.154	0.635	0.525	0.465	0.556	7.230	4.365
(1960–1991)					(0.427)	(0.470)	(0.801)	(0.695)	(0.008)	(0.001)
Japan	0.002	−1.042	1.719	0.328	6.116	3.345	4.238	4.728	5.453	2.476
(1970–1991)					(0.016)	(0.072)	(0.002)	(0.002)	(0.002)	(0.040)
United Kingdom	−0.009	−0.606	2.892	0.189	1.387	0.081	3.208	1.413	7.028	2.758
(1962–1991)					(0.242)	(0.777)	(0.010)	(0.235)	(0.009)	(0.022)

Table IV

Regression Analysis of the Effects of Oil Shocks on Real Stock Returns Conditional on Cash Flows *and* Expected Returns Using Quarterly Data for the United States, Canada, Japan, and the United Kingdom.

This table contains estimates of regression (7):

$$\mathrm{RS}_t = \alpha_4 + \sum_{i=1}^{5} a_i\mathrm{X}_{it} + \sum_{s=0}^{4} \theta_s\mathrm{OIL}_{t-s} + \sum_{s=0}^{4} \beta_s\mathrm{IP}_{t+s} + \eta_{4t}$$

where RS_t = real stock return in quarter t measured as the difference between the continuously compounded nominal stock return, S_t, calculated from a country's stock market index and the inflation rate, I_t, computed as the logarithmic difference in the consumer price index (CPI); IP_t = growth rate of industrial production in quarter t calculated as the logarithmic difference in the (seasonally adjusted) index of industrial production; OIL_t = prewhitened percentage change in the oil price in quarter t computed as the logarithmic difference in the producer price indices for fuels and related products and power for the United States, for petroleum and coal products for Canada and Japan, and for fuel for the United Kingdom; and X_{it} = proxies for expected returns and shocks to expected returns, where X_{1t} = dividend yield measured at time $t-1$, DY_{t-1}, is the average of monthly dividend yields on the country's market index for the past twelve months, X_{2t} = default spread measured at time $t-1$, DEF_{t-1}, is the difference between the annual yields on portfolios of corporate bonds, CBY_{t-1}, and long-term government bonds, GBY_{t-1}, X_{3t} = term spread measured at time $t-1$, TERM_{t-1}, is the difference between the annual yields on a portfolio of long-term government bonds, GBY_{t-1}, and on the a short-term Treasury bill, TBY_{t-1}, X_{4t} = shocks to the default spread at time t, SDEF_t, is the residual from a first order autoregressive model for DEF_t, and X_{5t} = shock to the term spread at time t, STERM_t, is the residual from a first order autoregressive model for TERM_t. We report the intercepts and the *sum* of the coefficients of independent variables. For brevity, we do not report the coefficient of the expected return variables, X_{it}, but we do report the F-test (see H_1) for their joint significance. All hypotheses tests, H_1 through H_7, are conducted using standard F-statistics; the p-values of these statistics are reported in parentheses. The numbers in parentheses in the $\bar{R}^2$ column are the $\bar{R}^2$s of the regressions of RS_t on the expected return, X_{it}, and the real cash flow, IP_{t+s}, variables alone.

Country	α_4	$\sum_{s=0}^{4} \theta_s$	$\sum_{s=0}^{4} \beta_s$	$\bar{R}^2$	H_1: $a_i = 0\ \forall i$	H_2: $\sum_{s=0}^{4} \theta_s=0$	H_3: $\sum_{s=1}^{4} \theta_s=0$	H_4: $\theta_s = 0\ \forall s$	H_5: $\theta_s = 0\ \forall s > 0$	H_6: $\sum_{s=0}^{4} \beta_s=0$	H_7: $\beta_s = 0\ \forall s$
United States	−0.055	−0.351	1.279	0.189	1.478	0.923	0.367	0.759	0.609	7.866	3.978
(1947–1991)				(0.195)	(0.200)	(0.338)	(0.546)	(0.581)	(0.657)	(0.006)	(0.002)
Canada	−0.061	−0.853	1.359	0.206	2.410	3.174	2.945	0.850	1.005	2.053	2.477
(1960–1991)				(0.211)	(0.041)	(0.078)	(0.089)	(0.518)	(0.408)	(0.155)	(0.037)
Japan	0.033	−0.658	1.917	0.361	1.727	1.659	1.107	2.267	2.815	5.915	2.080
(1970–1991)				(0.303)	(0.141)	(0.202)	(0.297)	(0.058)	(0.032)	(0.018)	(0.079)
United Kingdom	−0.162	−0.755	3.494	0.236	2.240	2.079	0.008	3.245	1.825	8.567	2.548
(1962–1991)				(0.151)	(0.056)	(0.153)	(0.928)	(0.010)	(0.130)	(0.004)	(0.033)

Country	$\sum \theta_s$	$\sum \beta_s$	$\bar{R}^2$	no-oil $\bar{R}^2$	p-value: expected-return vars	p-value: all oil vars	p-value: all cash-flow vars
United States	-0.351	1.279	0.189	0.195	0.200	0.581	0.002
Canada	-0.853	1.359	0.206	0.211	0.041	0.518	0.037
Japan	-0.658	1.917	0.361	0.303	0.141	0.058	0.079
United Kingdom	-0.755	3.494	0.236	0.151	0.056	0.010	0.033

Read. For the United States and Canada, oil variables do not add meaningful explanatory power after cash-flow and expected-return variables are included. For Japan, the oil coefficient tests remain close to conventional significance levels, and adding oil raises adjusted R^2 from 0.303 to 0.361. For the United Kingdom, the oil variables remain significant, and adding oil raises adjusted R^2 from 0.151 to 0.236.

Economic message. Expected-return controls do not solve the Japan/U.K. puzzle. These markets still display oil-related return movements that are not explained by the measured rational channels.

5.6 Additional robustness evidence

Nominal returns. Repeating the analysis with nominal stock returns weakens some measured relations, but the qualitative pattern remains.

Oil-price measurement error. Classical measurement error in oil-price indices would bias oil effects toward zero, so it cannot easily explain the residual significance of oil shocks in Japan and the United Kingdom.

Cash-flow measurement error. Augmented regressions with future stock returns do not remove the Japan/U.K. residual oil effects. The required amount of cash-flow noise needed to rationalize the residual results is large.

Common sample. Reestimating on a common 1970–1991 period gives qualitatively similar results.

6 Short summary

Jones and Kaul ask whether international stock markets rationally process oil shocks. Their test is based on a simple valuation idea: oil shocks can move stock prices because they change expected real cash flows or expected returns. If those channels are controlled for, oil shocks should have no separate effect.

The paper first establishes that real stock returns are related to current and future industrial production growth. It then shows that oil-price increases negatively affect stock returns in the United States, Canada, Japan, and the United Kingdom.

The central result comes from conditioning stock returns on cash-flow variables. In the United States and Canada, future real activity explains the stock-market reaction to oil shocks. This is consistent with rational valuation: oil prices reduce expected cash flows, and stock prices adjust accordingly.

Japan and the United Kingdom are different. In these countries, oil shocks continue to explain stock returns even after controlling for future industrial production growth. Adding expected-return proxies such as dividend yields, default spreads, term spreads, and spread shocks does not fully eliminate the oil effect.

The authors then examine measurement-error concerns. Inflation measurement error, oil-price measurement error, cash-flow measurement error, and sample-period differences do not overturn the main pattern. The evidence suggests that the U.S. and Canadian markets price oil shocks through fundamentals, while the Japanese and U.K. stock-market reactions appear excessive relative to measured fundamentals.

7 Conclusion

7.1 Core conclusion

The paper finds mixed cross-country evidence on stock-market rationality. U.S. and Canadian stock-price responses to postwar oil shocks can be accounted for by the effect of those shocks on real cash flows. Japanese and U.K. stock-price responses cannot be fully explained by measured cash-flow changes or by standard expected-return proxies.

7.2 Economic intuition

A rational equity market should ask: how does an oil shock change the present value of future cash flows? In the United States and Canada, the answer appears to match the later movement in real activity. In Japan and the United Kingdom, stock prices fall more than the measured real cash-flow response seems to justify.

7.3 Main empirical contribution

The paper contributes a clean empirical design for studying market rationality. Instead of relying only on financial predictors, it uses a real macro shock and real cash-flow proxies. This makes the test more directly tied to fundamental valuation.

7.4 What remains unresolved

The Japan/U.K. evidence does not prove irrationality beyond all doubt. It is possible that oil shocks change expected returns through mechanisms not captured by dividend yields, default spreads, term spreads, or spread shocks. But within the paper's measured rational channels, these two markets display excess sensitivity to oil shocks.

7.5 Takeaway

Oil shocks provide a useful test of whether stock markets translate macroeconomic news into prices through fundamentals. Jones and Kaul show that this translation works well for the United States and Canada, but less well for Japan and the United Kingdom.